Aaradhya Dev Tamrakar

AI Systems Engineering · Distributed Orchestration · Developer Tooling

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Final-year engineering student building autonomous multi-agent developer tooling, distributed orchestration harnesses, and high-performance system utilities. Architect of Fleet-Orchestrator (multi-account agent fleet with SQLite WAL lease machines), custom MCP servers (NotebookLM, LocalSend, CAD), and cross-IDE configuration synchronizers.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Engineered a five-stage agentic Text-to-SQL pipeline (Planner → Generator → Validator → Executor → Summarizer), achieving **100% execution success and answer accuracy** on a 50-question project benchmark.
- Built containerized AI microservices and evaluation APIs using **FastAPI**, **PostgreSQL**, and **Docker**.

NSSR DataCamp Fellow (Cohort 2) — Nepalese Society of Student Researchers | April 2026 – Present

- Selected in competitive cohort for sponsored DataCamp access; developed six applied GenAI solutions automating data cleaning, schema normalization, and null filtering.

RESEARCH & TECHNICAL PROJECTS

Headless Invariant Assurance Engine (INV-BMK-001) | 2026

Python · Z3 · SMT-LIB2 · State Machines · Pytest · Formal Methods

- Developed a formal verification benchmark encoding reactive state-machine contracts into **SMT-LIB2/Z3**, inductively proving candidate safety invariants and generating executable counterexamples.

Fleet-Orchestrator — Multi-Account Autonomous Agent Swarm & Consensus Engine | 2026 – Present

Python 3.12+ · FastAPI · SQLite WAL · asyncio · MCP · CDP · Copilot CLI

- Architected an autonomous multi-agent coordinator managing background worker swarms across **31 Copilot accounts** with atomic **SQLite WAL lease state machines** and crash recovery.
- Implemented multi-provider adapters (Copilot CLI, Gemini, Claude, local SLMs), quota-aware round-robin scheduling, and adversarial lease-stealing tests.

Super-NLM — Multi-Account NotebookLM Aggregator & MCP Hub | 2026

Python · FastAPI · React · MCP · asyncio

- Developed an MCP stdio/HTTP server and dashboard enabling multi-account NotebookLM access with round-robin rotation, quota-aware backoff, and cross-notebook synthesis.
- Automated Google Drive folder synchronization and batch generation of Studio artifacts (audio overviews, study guides, mind maps).

Agent-Customization-Sync — Cross-IDE Configuration Synchronizer | 2026

Python · Win32 NTFS Junctions · SHA-256 · Pytest

- Engineered cross-IDE configuration synchronizer providing snapshotting, rules transpilation, and **SHA-256 drift auditing** across Antigravity, Cursor, and Claude Code.

Nexus — Personal AI Workspace & Knowledge Retrieval Hub | 2025 – 2026

FastAPI · React (Vite) · SQLite FTS5 · asyncio · TailwindCSS

- Architected a local AI workspace integrating multi-model API backends with **SQLite FTS5 full-text search** for sub-15ms semantic note retrieval.
- Engineered responsive frontend with React, Vite, and WebSocket streaming for low-latency conversational interaction.

Win-Governor / System-Optimizer — Workstation Memory Reclamation Engine | 2025 – 2026

C# .NET 10 · WPF · Win32/NT APIs · Windows Internals

- Built a native Windows performance utility reclaiming **1.2–3.4 GB working-set RAM** under heavy developer workloads via undocumented Win32/NT APIs.

Utility MCP Suite — LocalSend, Google Classroom & Fusion 360 Bridges | 2026 – Present

Node.js ESM · Python · MCP Protocol · REST / HTTP · mTLS

- Developed **localsend-mcp** enabling AI agents to discover LAN peers and transfer multi-gigabyte build artifacts over secure mTLS (<25ms transport latency).
- Built **fusion360-mcp** establishing a headless IPC bridge to Autodesk Fusion 360 for agentic 3D parametric CAD modeling.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Direct branch strategy, executive committee operations, technical workshops, and peer learning initiatives for engineering students.
- Previously managed operational documentation and workshop logistics across department-level technical events as Vice Secretary.

Makerspace Ambassador — KEC Maker's Space, Kathmandu Engineering College | June 2026 – Present

- Support student rapid prototyping, 3D printing, and PCB bring-up; maintain lab website infrastructure and asset consistency.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL (PostgreSQL, SQLite), Kotlin, C# (.NET), VHDL, PowerShell, Bash, JavaScript / TypeScript

AI Systems, Backend & MLOps: FastAPI, Docker, Model Context Protocol (MCP), asyncio, MLflow, Evidently AI, Airflow, PostgreSQL, SQLite FTS5, REST APIs

Engineering Practice & Tools: Git, GitHub, Linux (Ubuntu), Jupyter, Google Colab, LaTeX / Overleaf, WordCraft, Pytest, CMake / CTest, Automated CI/CD